Presentation

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Section 1

INTRODUCTION

Beef industry context

- Beef palatability depends on tenderness, juiciness, and flavor.
- In the United States, there is a quality grading system that classifies beef carcasses on different quality categories (USDA) based on animal maturity and marbling level (intramuscular fat).
- Tenderness is not predicted with USDA quality grade.
- Although some technologies allow the determination of beef tenderness, they require sacrificing a loin steak and cannot be performed in real-time.

Rapid Evaporative Ionization Mass Spectrometry (REIMS)

- Rapid Evaporative Ionization Mass Spectrometry (REIMS) is a novel environmental mass spectrometry technique that combines a sample vaporization device with a time-of-flight mass spectrometer.
- The sampling device, which uses a high-frequency current, comes into contact with the sample surface, an aerosol of molecules is generated and transformed into a mass spectral fingerprint.
- REIMS can classify biological tissues in-situ, in real-time and without any sample preparation.
- The goal of this study was to evaluate the ability of REIMS to predict USDA quality grade and beef tenderness.

Section 2

METHODOLOGY

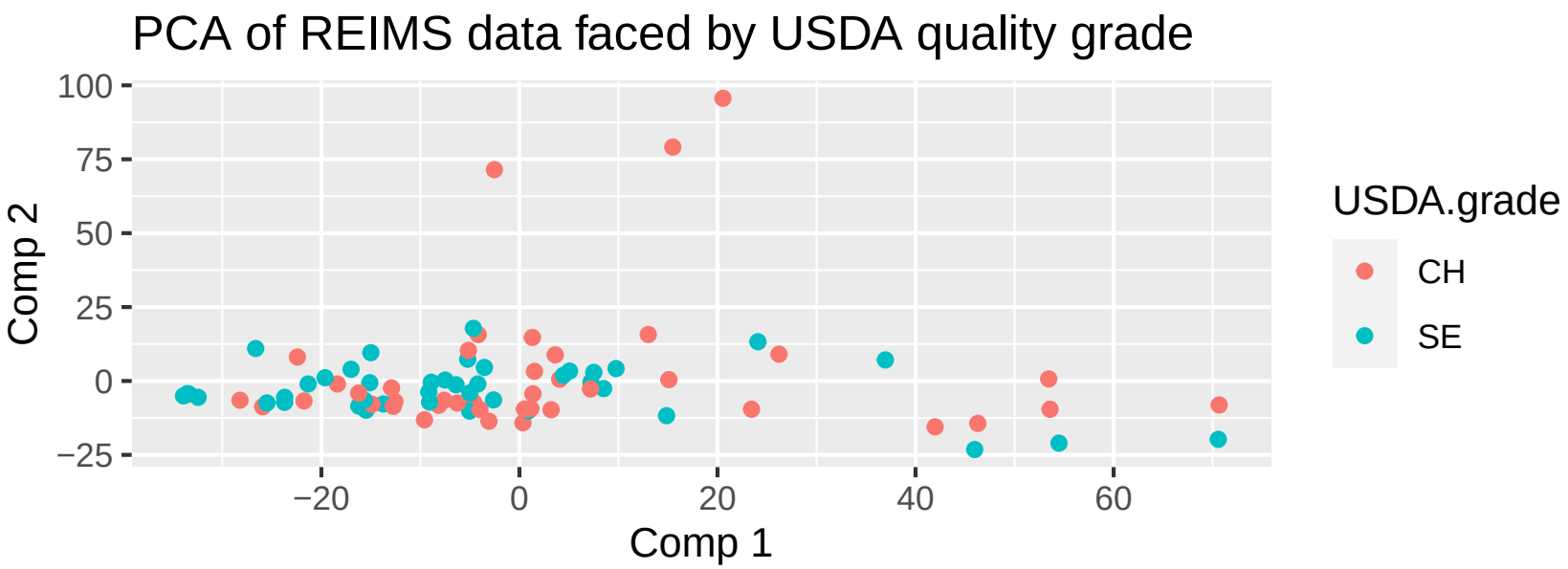
Data collection

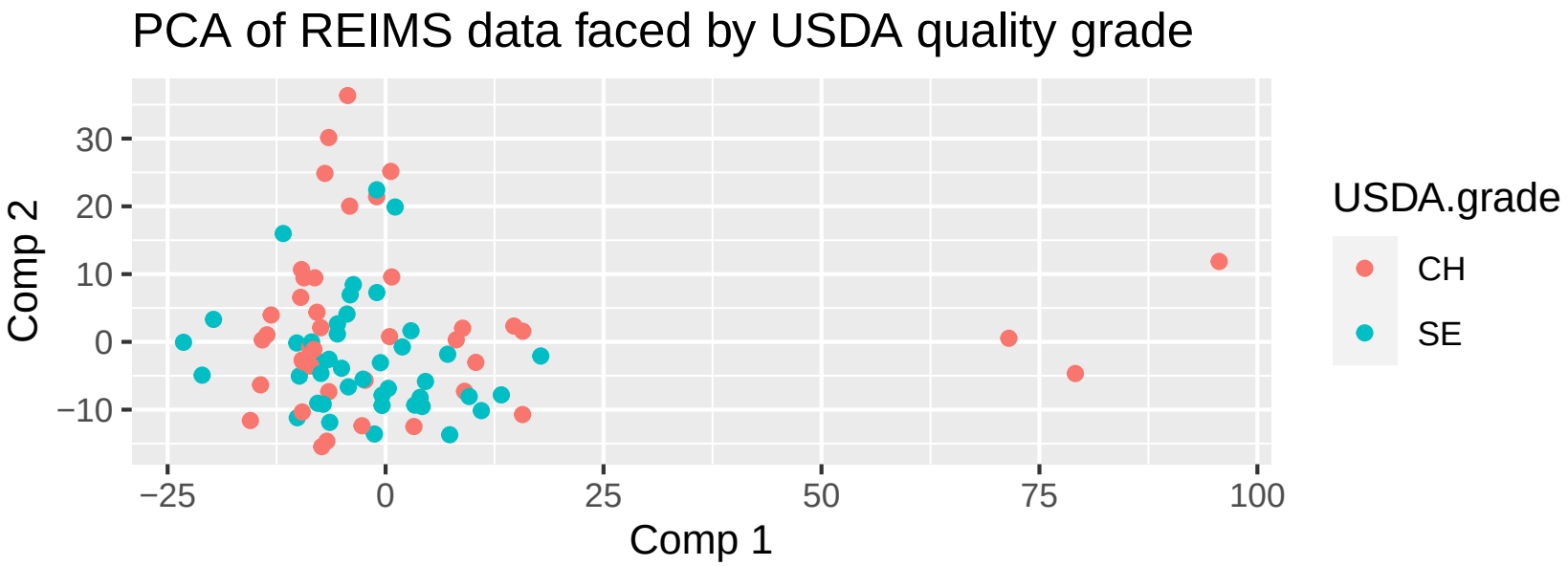
- Eighty-three whole striploins were collected at 36h post-mortem (42 USDA quality grade Choice, and 41 USDA quality grade Select).
- They were cut into three sections, which were aged at 3 different aging treatments (3 days, 14 days, 28 days).
- Tenderness were measure at each aging period.
- The number of observations is 83, and the number of predictors is 1700.
- Classification models were performed to predict USDA quality grade.
- Regression models were performed to predict tenderness.

Section 3

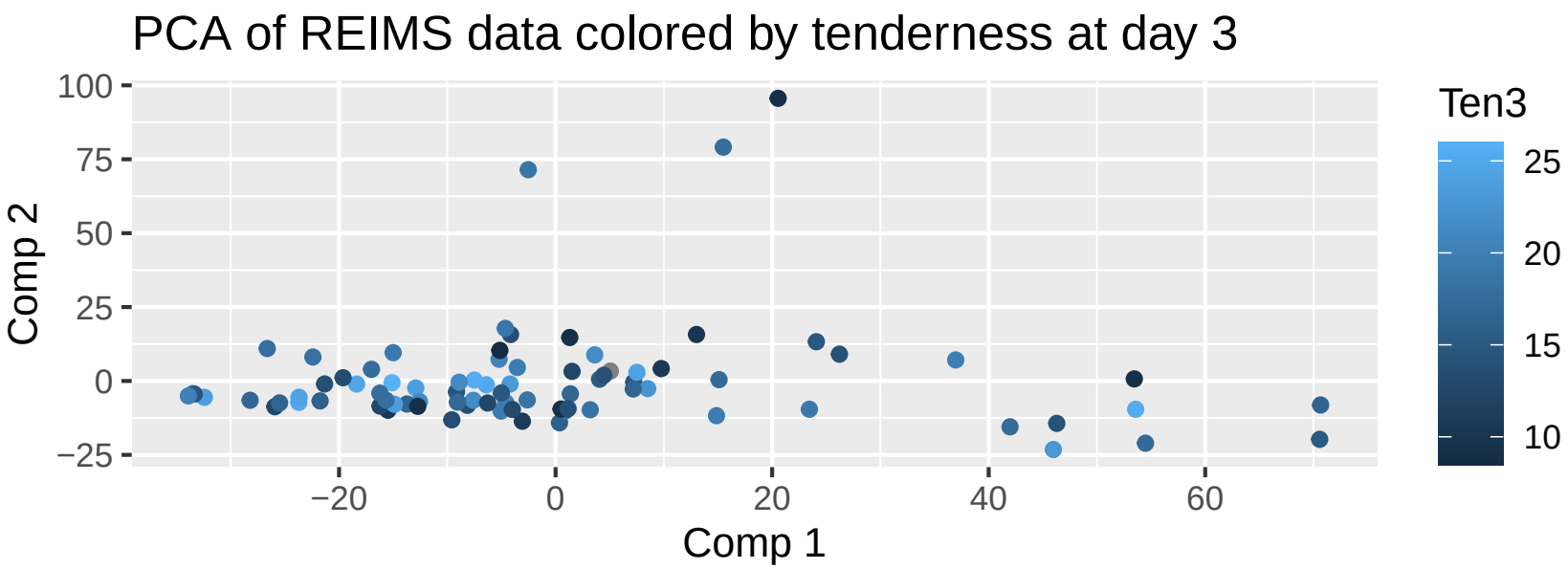
PCA

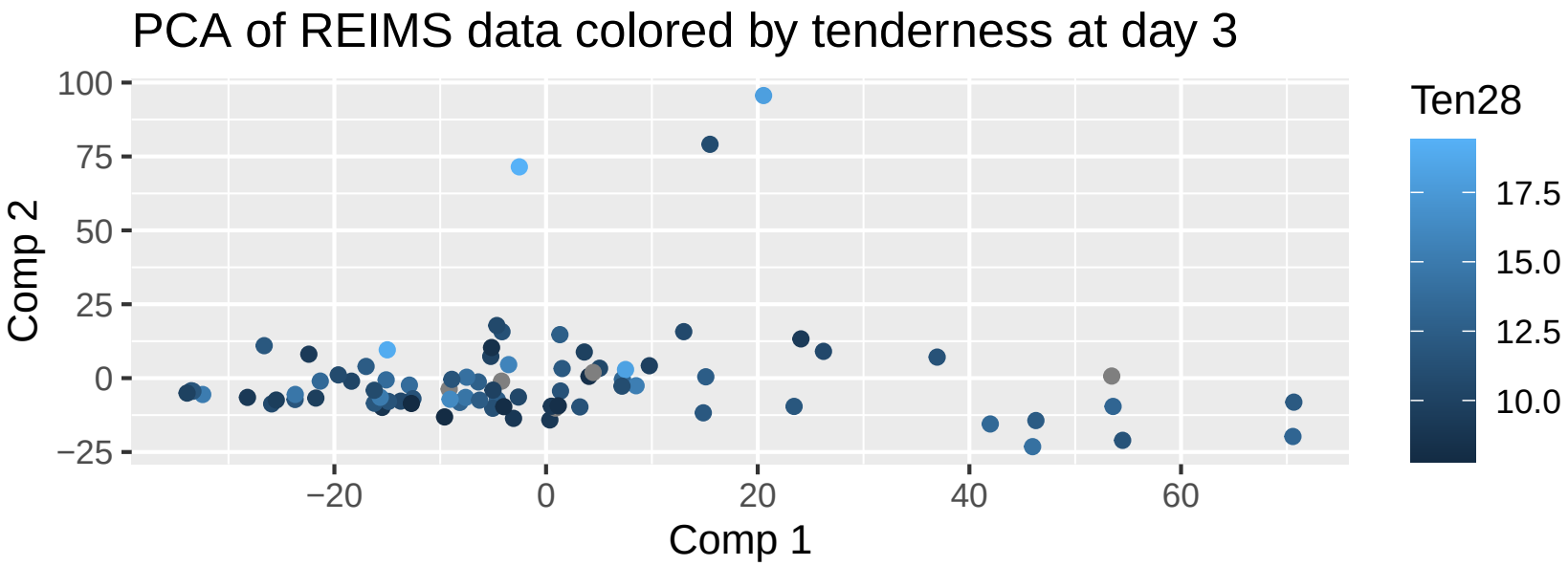
USDA quality grade



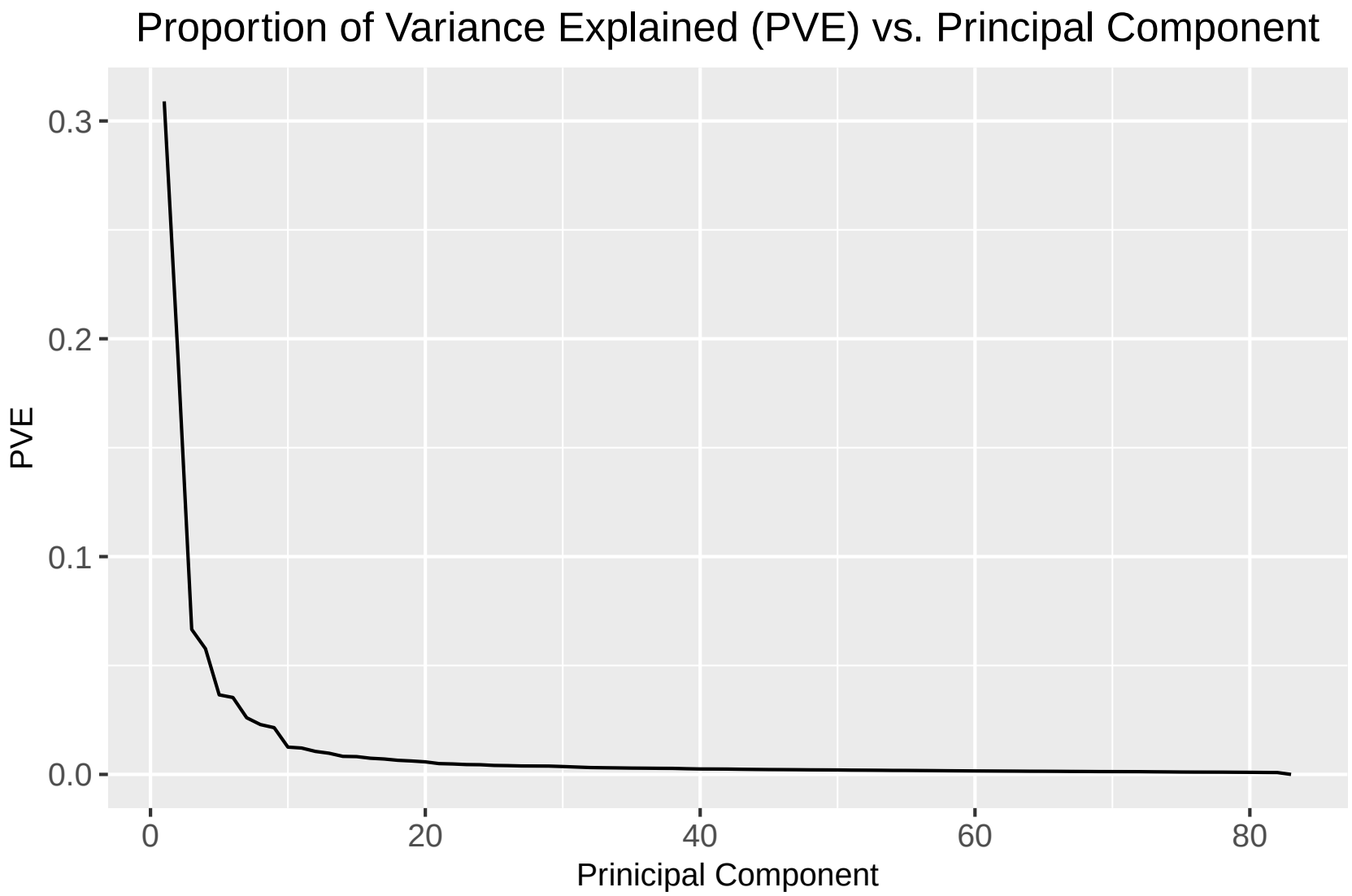


Tenderness


PCA of REIMS data colored by tenderness at day 3
Comp 2
Comp 1
100
75
50
25
0
−25
−20
0
20
40
60
Ten3
25
20
15
10


PCA of REIMS data colored by tenderness at day 3
Comp 2
Comp 1
100
75
50
25
0
−25
−20
0
20
40
60
Ten28
17.5
15.0
12.5
10.0

Proportion of Variance Explained



Section 4

CATEGORIAL MODELS: USDA GRADE

Methodology

- Several models were built to predict USDA Grade using the principal components:
  - Logistic Regression
  - Support Vector Machines (Linear, Polynomial, Radial)
- 10-Fold cross validation was performed on each model using an increasing number of principal components
- Optimal CV Error was recorded for each model and this was used to identify the optimal number of principal components to utilize for each model.

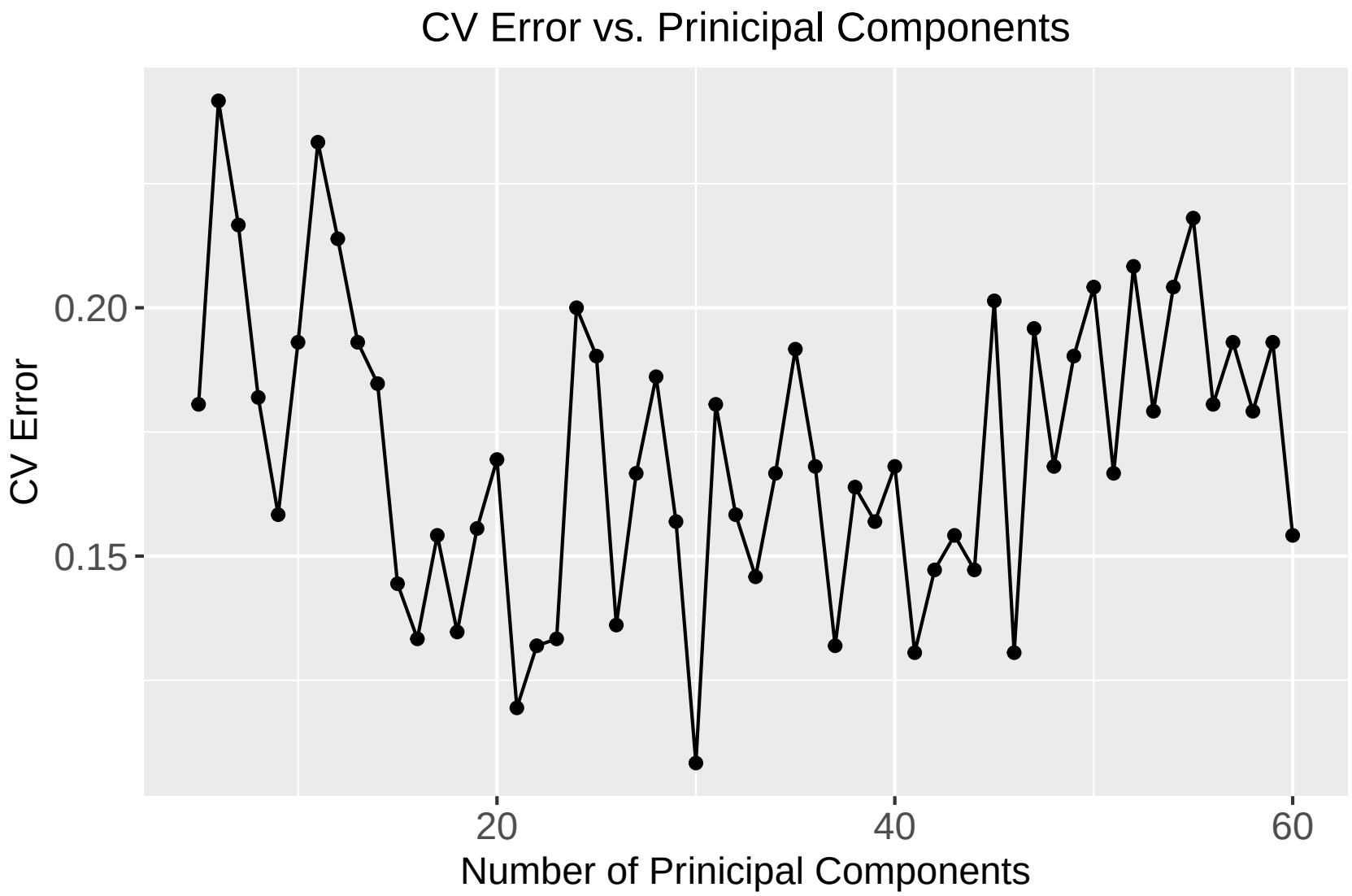
Radial Support Vector Machine

- Support Vector Machine with a Radial kernel.
- Built models using an increasing number of principal components from 2 to 60.
- For each iteration of principal components, a SVM was tuned using:
  - Costs from .0001 to 10,000
  - γ from .0001 to 10,000

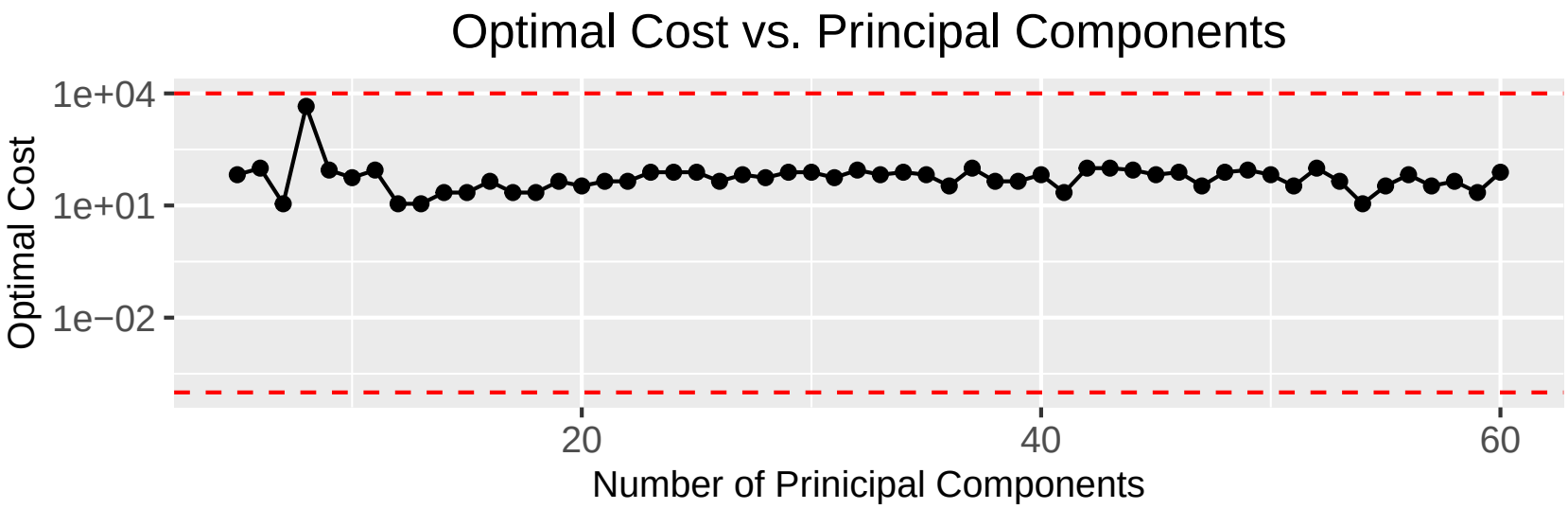
Table 1: Best performing Radial SVM Model.

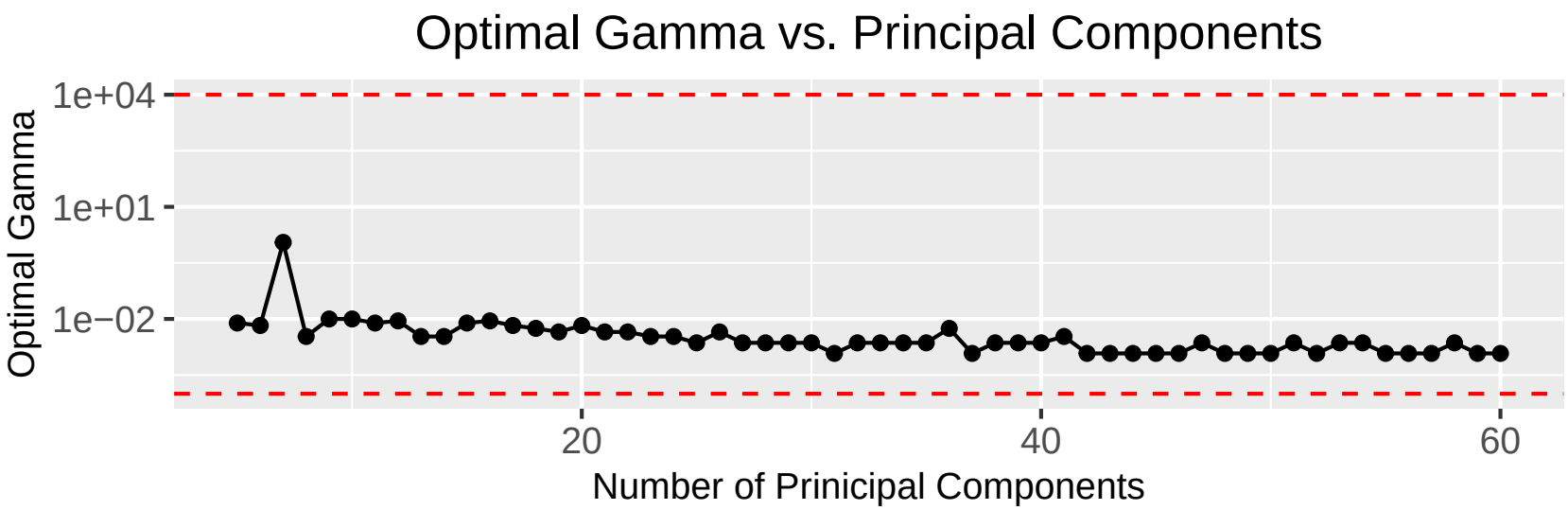
PC	CV_Error	Cost	Gamma	SV
30	0.108	77.78	0.0023	45

Radial Support Vector Machine



Radial SVM: Optimal Parameters





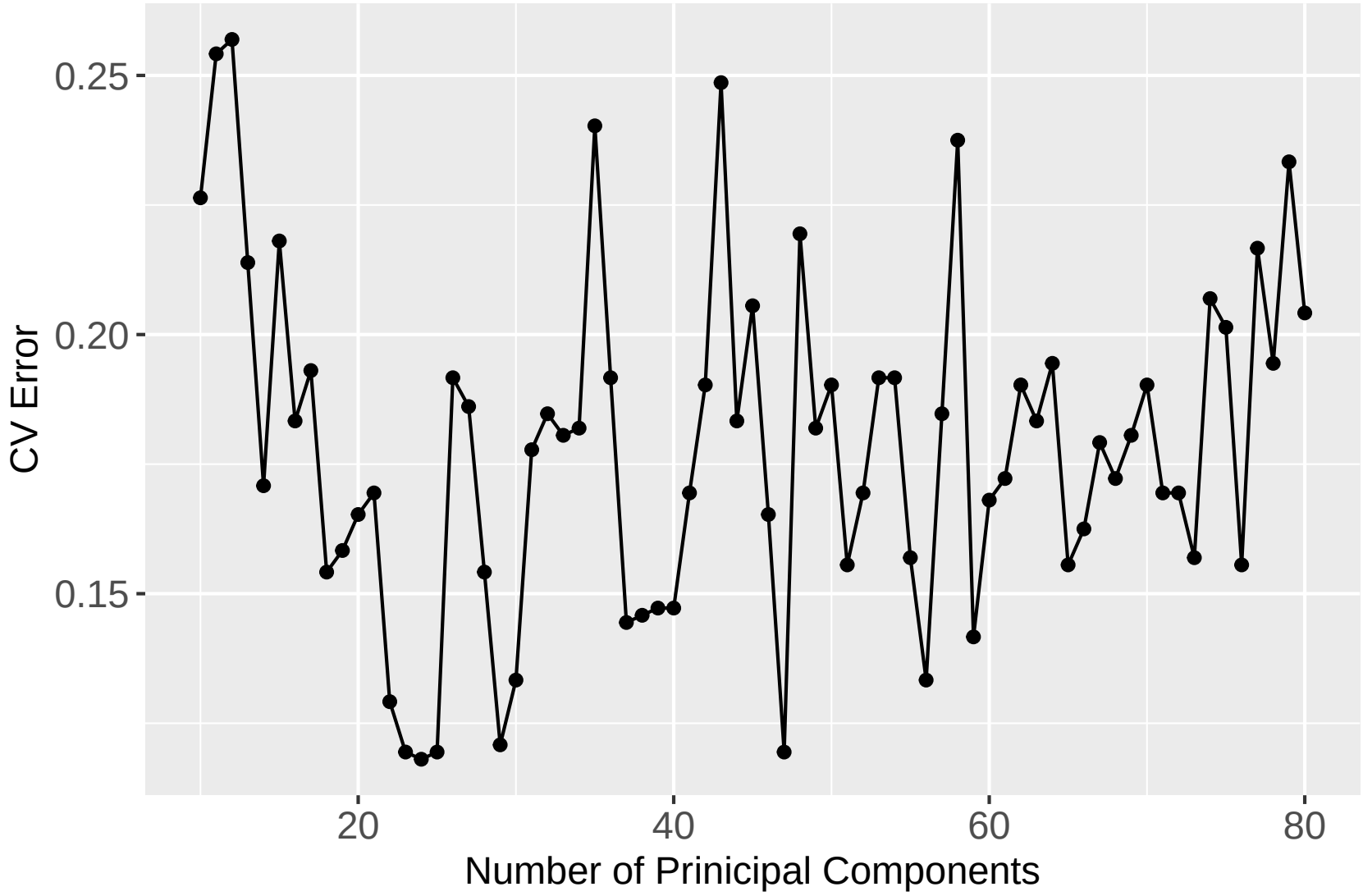
Linear Support Vector Machine

- Support Vector Machine with a Linear kernel.
- Built models using an increasing number of principal components from 2 to 80.
- For each iteration of principal components, a SVM was tuned using:
  - Costs from .01 to 100

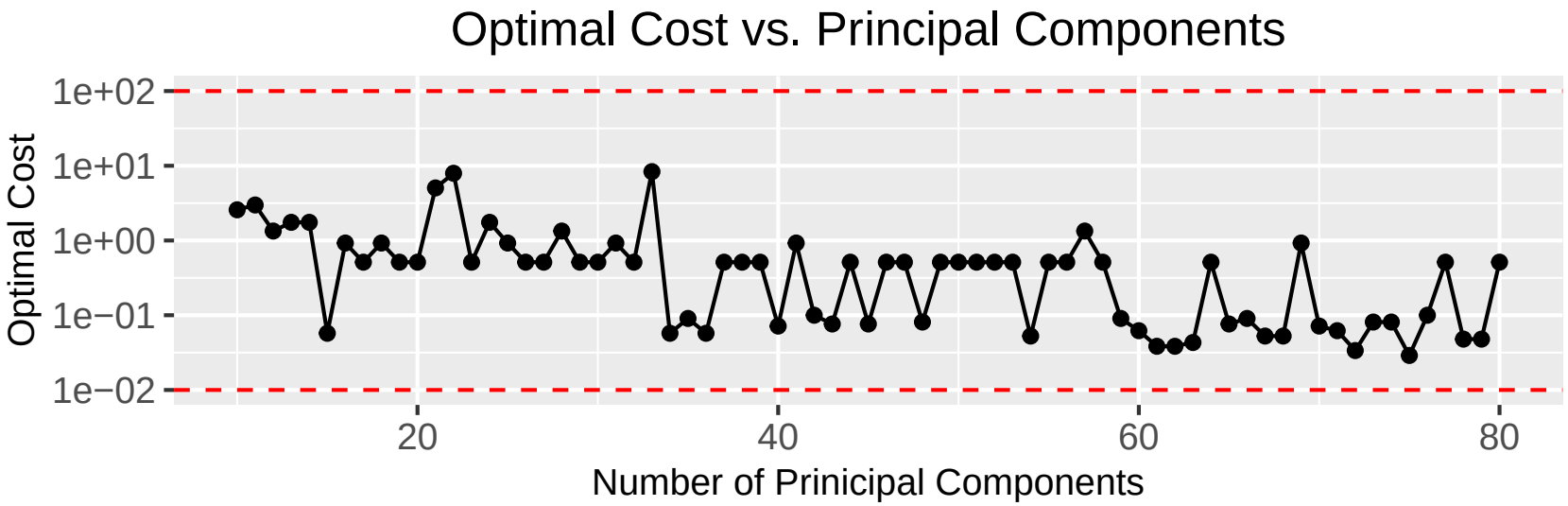
Table 2: Best performing Linear SVM Model.

PC	CV_Error	Cost	SV
24	0.118	1.75	32

Linear Support Vector Machine



Linear SVM: Optimal Parameters



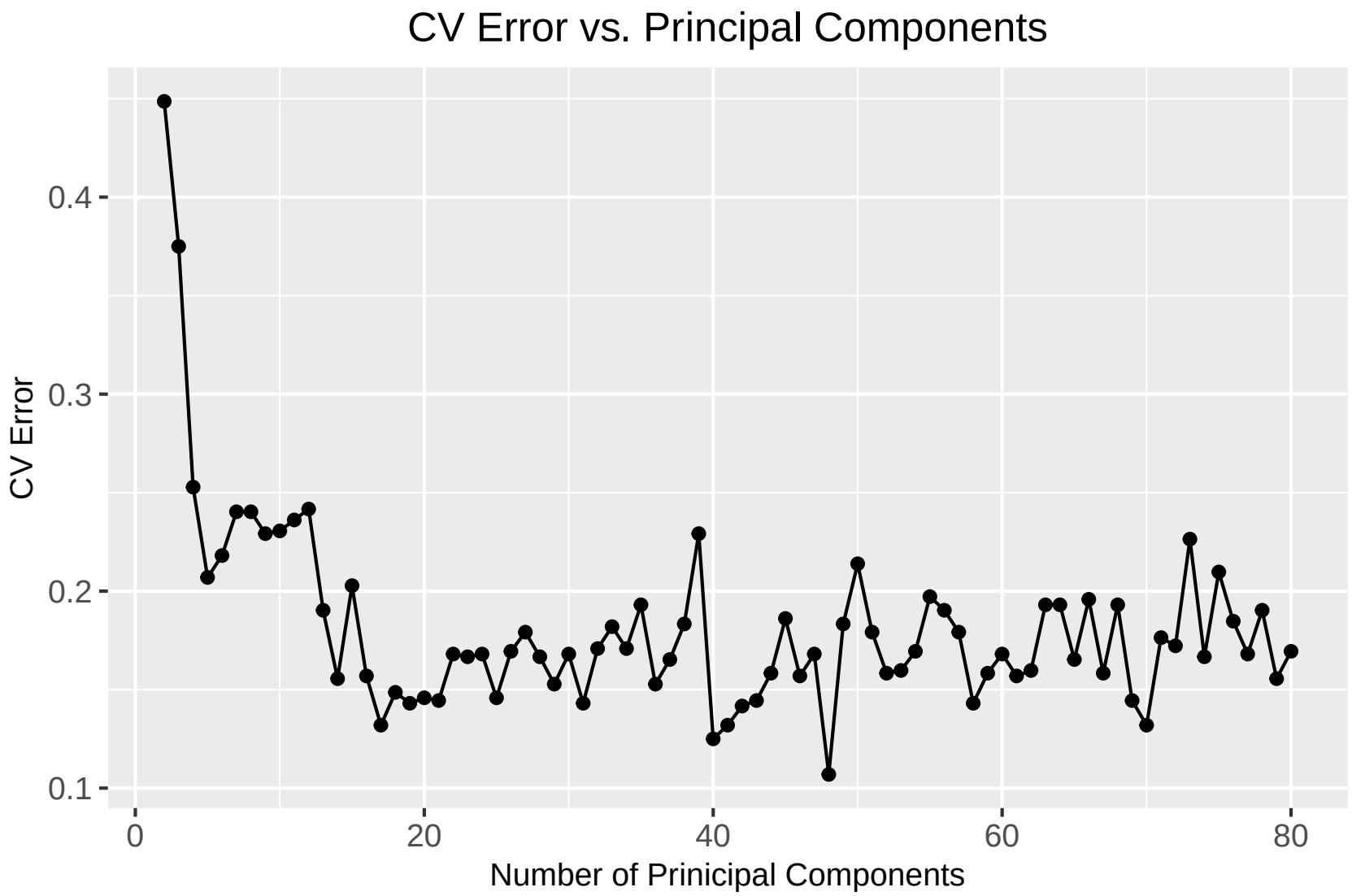
Polynomial Support Vector Machine

- Support Vector Machine with a Polynomal kernel.
- Built models using an increasing number of principal components from 2 to 60.
- For each iteration of principal components, a SVM was tuned using:
  - Costs from .01 to 100
  - Degrees from 1 to 4

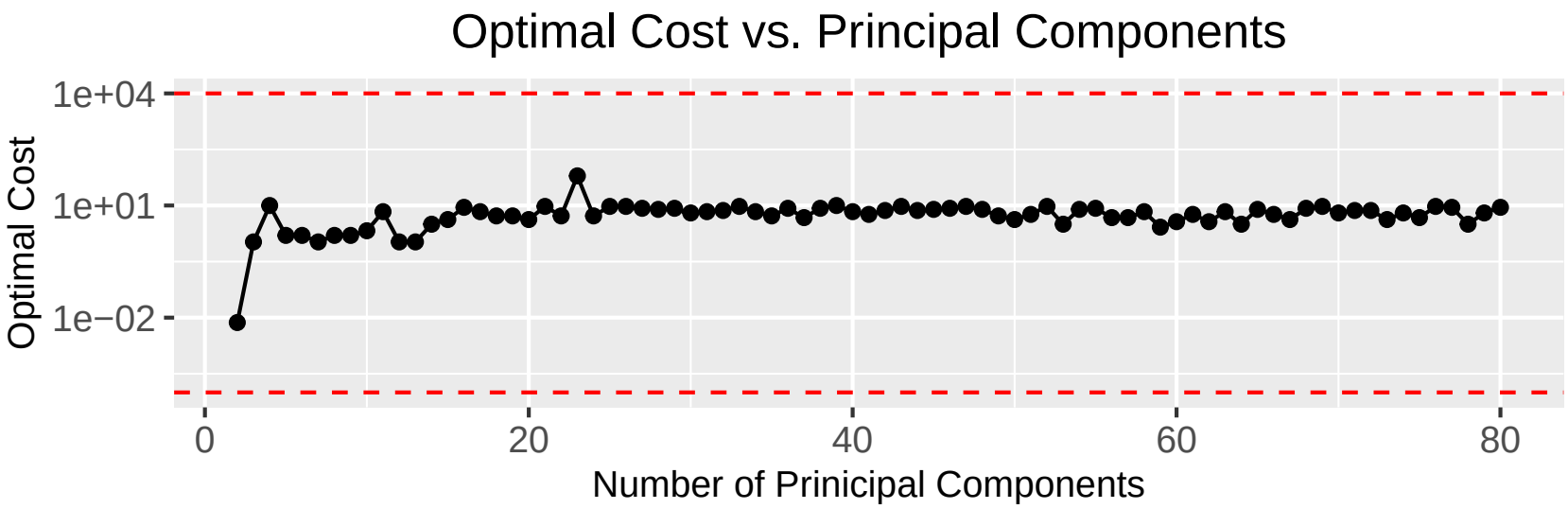
Table 3: Best performing Linear SVM Model.

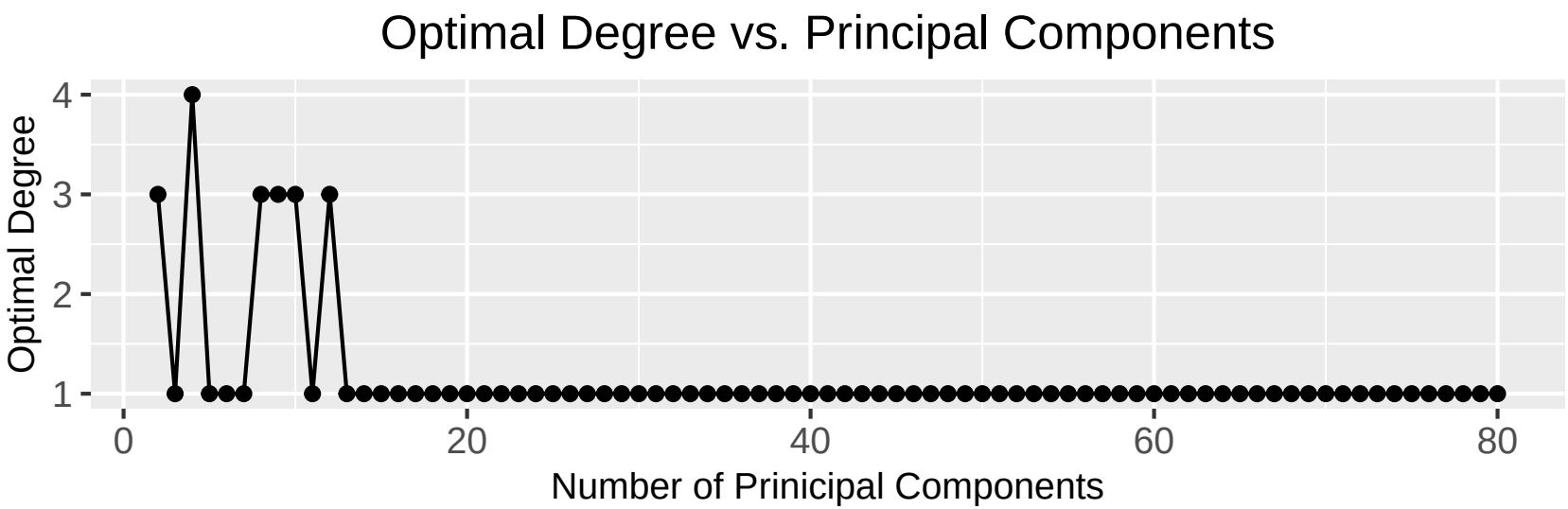
PC	CV_Error	Cost	Degree	SV
48	0.107	7.8968	1	46

Polynomial Support Vector Machine



Polynomial SVM: Optimal Parameters

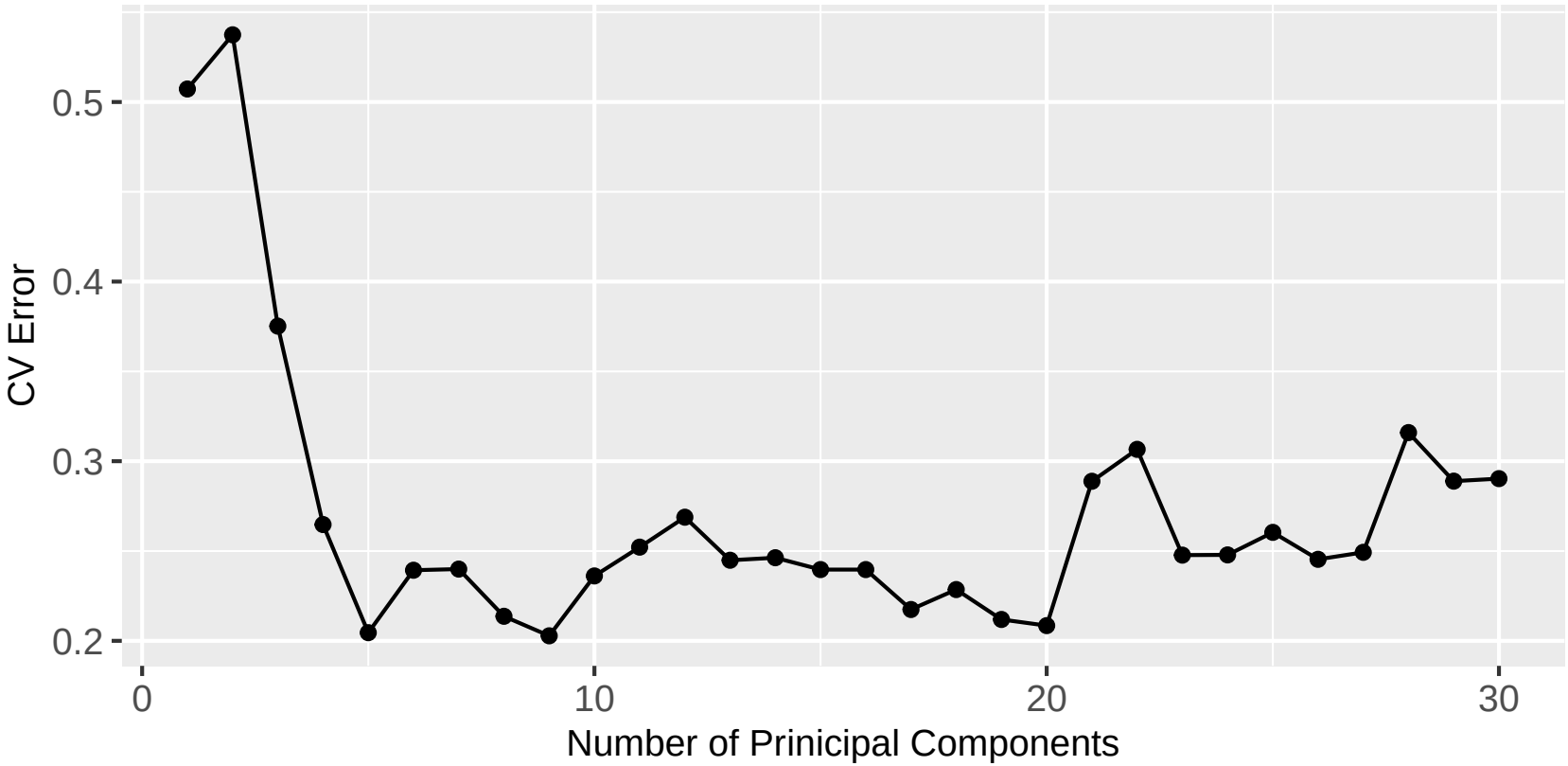




Logistic Regression Model

Table 4: Best performing logistic regression model.

PrincipalComponents	CV_Error
9	0.2



Summary of Categorical Model Results

Table 5: Best performing USDA Grade Models.

Model	PrincipalComponents	CV_Error
Radial SVM	30	0.108
Linear SVM	24	0.118
Polynomial SVM	48	0.107
Logistic Regression	9	0.203

- Best Model: Radial SVM with 30 Principal Components: CV Error of .108
  - Even though the Polynomial SVM outperformed the Radial SVM slightly, it required 48 more Principal Components to be utilized.
  - The Linear SVM performed slightly worse than the Radial SVM, but required 6 less Principal Components.

Section 5

REGRESSION MODELS: TENDERNESS

Summary results

Table 6: Mean Squared Error (MSE) of regression models for tenderness at day 3, 14, and 28

Model	Size	MSE_day3	Size.1	MSE_day14	Size.2	MSE_day28
Best Subset	1	23.87	1	9.93	1	7.98
PCA-Lasso	18	17.93	10	12.56	58	6.30
PCR	6	20.55	1	10.59	2	5.83
PCA-Splines	5	17.22	2	15.38	4	5.31